

A Review on Amino Acids

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Abstract

Protein is made up of amino acids, which are the building blocks of the molecule. Only 20 amino acids are involved in protein synthesis, even though more than 300 amino acids have been identified. In nature, all twenty amino acids did not appear at the same time. Rather, some of them appeared first, while others were later incorporated into the genetic code. It is vital to include them in one's diet since a lack of them causes a reduction in protein synthesis, which leads to disease. In this article, amino acids, their properties, functions, and associated diseases have been elaborated.

Keywords: Amino acids, protein, activity, diseases

INTRODUCTION

Amino acids are the building blocks of proteins. Amino acids are a class of chemicals that contain two functional groups: Amino (NH₂) and Carboxyl (COOH) (-COOH). The amino (-NH₂) group is basic, whereas the carboxyl (-COOH) group is acidic [1–6, 7]. Multiple processes associated with gene expression are regulated by amino acids, including the control of the function of proteins that mediate messenger RNA (mRNA) translation [7]. Acidic, basic, and neutral amino acids are the three types of amino acids. Because some amino acids are not produced by the body, they must be consumed through the food. These amino acids are referred to as essential amino acids. Some amino acids are generated in the body and do not require supplementation; these amino acids are referred to as non-essential amino acids. Some amino acids are generated in the body, but production is insufficient; these amino acids are referred to as semi-essential amino acids [8].

GENERAL STRUCTURE OF AMINO ACID

Amino acid, any of a group of organic molecules that contains a basic amino group (-NH₂), an acidic carboxyl group (-COOH), and an organic R group (or side chain) that's unique to every amino acid. The term amino acid is brief for α -amino [alpha-amino] acid. Each molecule contains a central carbon (C) atom, called the α -carbon, to which both an amino and a carboxyl group are attached. The remaining two bonds of the α -carbon atom are generally satisfied by a hydrogen (H) atom and the R

group [1–6, 9]. The key elements of an amino acid are carbon (C), hydrogen (H), oxygen (O), and nitrogen (N), although other elements are found within the side chains of certain amino acids [9]. About 500 present amino acids are known (though only 20 appear within the genetic code) and may be classified in some ways [10]. They will be classified in step with the core structural-functional groups' locations as alpha- (α -), beta- (β -), gamma- (γ -) or delta- (δ -) amino acids [9].

PROPERTIES OF AMINO ACIDS [4, 11, 12]

Physical Properties of Amino Acids

Solubility

Most of the amino acids are usually soluble in water and insoluble in organic solvents.

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Melting Point

Amino acids are generally melted at a higher temperature often above 2000 C.

Taste

Amino acids are also sweet (Glycine, Alanine and Valine), tasteless (Leucine) or Bitter (Arginine & Isoleucine).

Optical Properties

Optical Properties: All amino acids except glycine possess optical isomers because of the presence of asymmetric α -carbon atoms and show optical activity.

Zwitterion and Isoelectric Point

Zwitterion and Isoelectric point: The name zwitter is derived from the German word which implies hybrid. Zwitter ion (or) dipolar ion is a hybrid molecule containing positive & negatively ionic groups. The proton shifts from the carboxyl group to the amino group of the self-molecule at normal pH cellular levels.

Titration Curve of Glycine

Glycine is optically inactive, the simplest amino acid because which don't have an asymmetric carbon atom. Acid-Base titration involves the gradual addition (or) removal of protons. It's three different stages when the Glycine undergoes acid-base titration.

Chemical Properties of Amino acids

Chemical reactions of amino acids due to carboxyl and amino groups.

Due to Carboxyl group**Decarboxylation**

The amino acids will undergo alpha Decarboxylation to make the corresponding amines.

Histidine $\rightarrow$ Histamine + CO_2

Tyrosine $\rightarrow$ Tyramine + CO_2

Tryptophan $\rightarrow$ Tryptamine + CO_2

Reaction with Alkalies (Salt formation)

Amino acids react with alkalies to make Salts.

Reaction with Alcohols (Esterification)

Amino acids are reacted with an alcohol to form, Ester.

Reaction with Ammonia:

The carboxyl group of dicarboxylic amino acids reacts with NH_3 to form Amides.

Aspartic acid + $\text{NH}_3 \rightarrow$ Asparagine

Glutamic acid + $\text{NH}_3 \rightarrow$ Glutamine

Due to Amino group**Reaction with Mineral acids (Salt formation)**

When the amino acids are treated with mineral acids (like HCl), it forms Acid Salts.

Due to Amino and Carboxyl Group**Ninhydrin reaction**

When Amino acids (or) Imino acid react with Ninhydrin molecule it gives Color. When it gives Purple color (Ruhemanns Complex) the Unknown sample is Amino acids (Which have primary amine NH_2) or it gives Yellow color the Unknown sample is Imino acid ($-\text{NH}-$).

Amino acid + Ninhydrin → Keto acid + NH₃ + CO₂ + Hydrindantin
Hydrindantin + NH₃ + Ninhydrin → Ruhemanns purple

L- AND D- AMINO ACIDS

Only L-amino acids are manufactured in cells and incorporated into proteins. Some D-amino acids are found within the cell walls of bacteria, but not in bacterial proteins. Glycine, the only amino acid, has no enantiomers because it has two hydrogen atoms attached to the central carbon atom. Only if all four attachments are different can enantiomers occur [1, 2].

ESSENTIAL, SEMI-ESSENTIAL AND NON-ESSENTIAL AMINO ACIDS

Essential Amino Acids

These are amino acids that are not produced by the body and must be obtained through food. They're also known as essential amino acids. Essential amino acid deficiency can induce a variety of disorders, including neurological breakdown, suppression of normal mental development, and even death in young animals. Eight amino acids are very important for humans.

- E.g. Valine, Leucine, Isoleucine, Threonine, Methionine, Phenylalanine, Tryptophan, and Lysine.
- An adequate amount of essential amino acids are required to maintain the proper nitrogen balance.

Semi-Essential Amino Acids

The body produces some of these amino acids, but not at a rate that is sufficient to meet the body's demands.

e.g.: Arginine and Histidine.

Non-Essential Amino Acids

These are the amino acids that the human body produces. The carbon skeletons of lipids and carbohydrates metabolism, as well as the transformation of necessary amino acids, provide these amino acids. 12 non-essential amino acids in humans are particularly important to their health. e.g.: Alanine, Arginine, Aspartic acid, Asparagine, Cystine, Cysteine, Glutamic acid, Glutamine, Glycine, Histidine, Proline, and Serine [1-5, 8].

CLASSIFICATION

Amino acids can be classified into 5 types [1-6, 13, 10]

- a. Classification based on the structure:
- b. Classification based on the position of-NH₂:
- c. Classification based on the Nutritional requirement:
- d. Classification based on the Metabolic Fates
- e. Classification based on polarity:
- f. Classification Based On the Structure:

A three-letter or one-letter symbol is allocated to each amino acid. The amino acids in protein structure are typically represented by these symbols. Proteins include 20 amino acids, which are split into the following classes.

Amino Acids with Aliphatic Side Chains

Monoamine monocarboxylic acids are what they're called. The simplest amino acids are glycine, alanine, valine, leucine, and isoleucine, which make up this category. Because the last three amino acids (leucine, isoleucine, and valine) have branched aliphatic side chains, they are known as branched-chain amino acids.

Hydroxyl Group-containing Amino Acids

Serine, threonine, and tyrosine are amino acids with hydroxyl groups. Because of its aromatic character, tyrosine is frequently classified as an aromatic amino acid.

Sulphur Containing Amino acids

The two amino acids absorbed during protein synthesis are cysteine with a sulfohydryl group and methionine with a thioether group.

Acidic Amino Acids and their Amides

Aspartic acid and glutamic acids are dicarboxylic monoamino acids while asparagine and glutamine are their respective amide derivatives.

Basic Amino Acids

The three amino acids lysine, arginine (with guanidino group) and histidine (with imidazole ring) are dibasic monocarboxylic acids. They are highly basic in character.

Aromatic Amino Acids

Aromatic amino acids include phenylalanine, tyrosine, and tryptophan (which have an indole ring). Apart from them, histidine may also fall under this category.

Imino Acids

An unusual amino acid is a proline, which has a Pyrrolidine ring. Instead of the amino group ($-NH_2$) found in other amino acids, it possesses an Imino group ($=NH$). As a result, proline is a γ -amino acid.

Heterocyclic Amino Acids

Histidine, Tryptophan, and Proline.

Classification Based on the Position of $-NH_2$

Amino acids can be classified into **three** types:

1. **α -amino acid:** The amino group attached to the next carbon of the carboxyl group is called α -amino acid. All naturally occurring amino acids are in α -L-amino acids.
2. **β -amino acid:** β -amino acid refers to the amino group attached to the third carbon (numbering from the Carboxyl group) of amino acid. For example, one of the end products of Pyrimidine catabolism is β -alanine.
3. **γ -amino acid:** γ -amino acid refers to the amino group connected to the amino acids fourth carbon (numbering from the Carboxyl group). e.g. GABA (Gamma Amino Butyric Acid).

Classification Based On the Nutritional Requirement

Based on Nutritional requirements, amino acids can be divided into **three** types.

1. Essential Amino acids (EAA)
2. Non-Essential Amino acids (NEAA)
3. Semi-Essential Amino acids (SEAA)

Essential Amino acids (EAA)

In the human body, some amino acids are not synthesized. It should be obtained from food. They are essential for the individual's appropriate development and maintenance [14].

e.g.: MET, VAL, PHE, (Or)

PRO, THR, TRP, LEU

M=Methionine A=Arginine T=Threonine Trp=Tryptophan V=Valine I=Isoleucine L=Leucine
P=Phenylalanine H=Histidine L=Lysine

Non-Essential Amino acids (NEAA)

To meet biological demands, the body can synthesis roughly 10 amino acids, thus they do not need to be eaten in the food. e.g.: Glycine, Alanine, Serine, Cysteine, Asparagine, Aspartic acid, Glutamic acid, Glutamine, Tyrosine, and Proline.

Semi-Essential Amino acid:

The amino acids histidine and arginine are semi-essential. They are required in the diet of growing children. However, they are not required for mature individuals.

Classification Based On the Metabolic Fates

Amino acids can be classified based on the metabolic fate into **three** groups:

1. Purely ketogenic amino acids
2. Ketogenic and Glucogenic amino acids
3. Purely Glucogenic amino acids

Purely ketogenic amino acids

These amino acids can be used to make fat. Because it is transformed into ketone bodies, leucine and lysine are completely ketogenic.

Ketogenic and Glucogenic amino acids:

Part of the carbon skeleton of these amino acids will enter the ketogenic pathway during metabolism, while the other will enter the glucogenic pathway. e.g.: Lysine, Isoleucine, Phenylalanine, Tyrosine & Tryptophan are partially ketogenic and partially glucogenic.

Purely Glucogenic amino acids

These amino acids can be used as a starting point for the production of glucose or glycogen. Because they solely enter the glucogenic pathway, the remaining 14 amino acids are all fully glucogenic.

Classification Based On Polarity:

Amino acids can be classified based on their polarity into **four** groups:

1. Non-polar amino acids
2. Polar amino acids with no charge on 'R' group
3. Polar amino acids with positive 'R' group
4. Polar amino acids with negative 'R' group

Non-polar amino acids

These amino acids are also known as hydrophobic amino acids (water-hating) they have no charge on the 'R' group. e.g.: Alanine, Leucine, Isoleucine, Valine, Methionine, Phenylalanine, Tryptophan, and Proline.

Polar amino acids with no charge on 'R' group:

As a result, the 'R' group of these amino acids is uncharged. They do, however, have groups like hydroxyl, sulfhydryl, and amide, and they contribute to protein structure hydrogen bonding. e.g.: Glycine, Serine, Threonine, Cysteine, Glutamine, Asparagine, and Tyrosine.

Polar amino acids with positive 'R' group:

e.g.: Lysine, Arginine, and Histidine.

Polar amino acids with negative 'R' group:

The aromatic dicarboxylic monoamino acids-e.g.: Aspartic acid and Glutamic acid.

GENERAL SYNTHETIC METHODS FOR THE PREPARATION OF AMINO ACIDS

Strecker's Synthesis

An aldehyde or ketone is treated with a solution of ammonium chloride and potassium cyanide in this procedure (ammonium cyanide). When the product is hydrolyzed, the corresponding amino acid is produced. The nucleophilic addition of ammonia to aldehyde starts the reaction. Alpha amino acids

are produced by removing water and then adding cyanide to aldimine. The method is useful for preparing Glycine, Alanine, Serine, Valine, Methionine, Glutamic acid, Leucine, Isoleucine, Norleucine, and Phenylalanine [13, 15, 16].

Gabriel Phthalimide Synthesis

This approach involves treating an alpha halogenated acid ester with potassium phthalimide to produce a substituted phthalimide, which when hydrolyzed yields phthalic acid and amino acid [11, 17, 18].

AMINO ACIDS

Glycine

Glycine is used to treat schizophrenia, stroke, sleep issues, BPH, metabolic syndrome, and a few uncommon inherited metabolic illnesses.

Alanine

Alanine is utilized to boost immunity and provide the brain and central nervous system more energy [15].

Valine

L-valine is most commonly used in health supplements for its involvement in muscle synthesis and maintenance. Stress management, child growth, and immune system support are some of the other benefits of L-valine.

Leucine

It Increases the physical performance or strength in the short term. Also increases long-term physical performance or preventing muscle loss in a variety of conditions, including immobility. Lowering blood glucose levels in type 2 diabetes.

Isoleucine

It is required for children's optimum growth and the maintenance of nitrogen balance in adults.

Norleucine

Norleucine is an artificial amino acid that has been used to research protein structure and function in the lab. It has a similar structure to METHIONINE.

Phenylalanine

Depression, Parkinson's illness, multiple sclerosis, pain, acupuncture anesthesia, and osteoarthritis are all treated with phenylalanine.

Tyrosine

Tyrosine is commonly utilized in the production of thyroid hormone. Tyrosine is a supplement that helps with mental performance, alertness, and memory.

Serine

Serine is particularly significant for the brain and central nervous system to function correctly.

Cysteine

In medicine, cysteine is used to boost the efficiency of corticosteroid medications.

Cystine

L-Cystine can be used in the treatment of surgery and trauma.

Threonine

Spinal spasticity, multiple sclerosis, familial spastic paraparesis, and amyotrophic lateral sclerosis are all conditions that are treated with threonine.

Methionine

Methionine is used to prevent and cure a variety of ailments, including depression, inflammation, liver illness, and muscle discomfort.

Iodogorgic acid

Diiodothyronine is a hormone. It's a type of medicine. Diiodothyronine is used to help people reduce weight, lower cholesterol, and develop muscle.

Thyroxine

Thyroid underactivity is treated with levothyroxine (hypothyroidism). It replaces or increases the amount of thyroid hormone generated by the thyroid gland.

Tryptophan

Tryptophan has been used to treat anxiety and depression in the past. It is used to treat migraines, headaches, and to boost the production of growth hormones.

Proline

One of the most important reasons to use L-proline supplements is to improve joint and skin health. It could also help with muscle development and injury healing.

Hydroxyproline

Hydroxyproline is required for the formation of collagen, the body's principal structural protein.

Aspartic acid

It's occasionally used to treat stress and exhaustion. It aids in the generation of energy from carbohydrates as well as the manufacturing of RNA and DNA.

Asparagine

It is necessary for brain development and function.

Glutamic acid

Glutamic acid serves as a source of energy for the brain. It improves brain clarity, mental alertness, and mood. As a result, it's utilized to treat Parkinson's, Fatigue, and Schizophrenia.

Beta-hydroxy glutamic acid

It is used to treat acne, photoaging, dry skin, psoriasis, and melasma, among other dermatological diseases. They've also been studied for fibromyalgia, dental scaling, and dry mouth, as well as the prevention and treatment of urinary tract stones.

Glutamine

It's used to keep blood sugar levels under control and consistent.

Ornithine

L-ornithine is primarily used in health supplements to support sports and performance. It's also utilized to help with liver function and wound healing.

Arginine

Because of its antiatherogenic, anti-ischemic, antiplatelet, and antithrombotic effects, L-arginine is a non-essential amino acid that may play an important role in the treatment of cardiovascular disease.

Lysine

Lysine appears to minimize cold sores when taken orally and when applied to the skin as a cream.

Histidine

Hemoglobin has a high percentage of histidine. As a result, it is used to treat anemia. It can reduce blood pressure. In baby feeding, histidine is critical.

AMINO ACIDS ARE USEFUL AS DRUGS

Certain non-standard Amino acids are used as drugs (Table 1) [1]:

- **D-Penicillamine:** It is a metabolite of penicillin. It is employed in the chelation therapy of Wilson's disease.
- **N-Acetylcysteine:** It is used in cystic fibrosis, and chronic renal insufficiency, as it can function as an antioxidant.
- **Gabapentin:** It is used as anticonvulsant.

Table 1. Overview of amino acids.

Amino Acids	Chemical Name	Three-letter abbreviation	One-letter abbreviation
Glycine	Amino acetic acid	Gly	G
Alanine	Alpha Aminopropionic acid	Ala	A
Valine	alpha-Aminoisovaleric acid	Val	V
Leucine	alpha-Aminoisocaproic acid	Leu	L
Isoleucine	alpha-Amino-beta-methyl n-valeric acid	Ile	I
Norleucine	alpha-Amino n-caproic acid	Nle	-
Phenylalanine	alpha-Amino-beta-phenyl propionic acid	Phe	F
Tyrosine	alpha-Amino-beta-(p-hydroxyphenyl) propionic acid	Tyr	Y
Serine	alpha-Amino-beta-hydroxypropionic acid	Ser	S
Cysteine	alpha-Amino-beta-mercaptopropionic acid	Csy	C
Threonine	alpha-Amino-beta-hydroxy-n-butyric acid	Thr	T
Methionine	alpha-Amino-gamma-methyl-thio-n-butyric acid	Met	M
Tryptophan	alpha-Amino-beta-indolepropionic acid	Trp	W
Proline	Pyrrolidine-alpha-carboxylic acid	Pro	P
Hydroxyproline	Gamma-Hydroxypyrrolidine alpha-carboxylic acid	Hyp	-
Aspartic Acid	alpha-Aminosuccinic acid	Asp	D
Asparagine	alpha-Aminosuccinamic acid	Asn	N
Glutamic Acid	alpha-Aminoglutaric acid	Glu	E
Beta-Hydroxy Glutamic Acid	Alpha-Amino-beta-hydroxyglutaric acid	Glx	Z
Glutamine	Alpha-Amino glutamic acid	Gln	Q
Ornithine	alpha-delta-Diamino-n-valeric acid	Orn	O
Arginine	alpha-Amino-delta-guanidino-n-valeric acid	Arg	R
Lysine	alpha-Diaminocaproic acid	Lys	K
Histidine	alpha-Amino-beta-imidazolepropionic acid	His	H

CONCLUSIONS

Amino acids are necessary for protein synthesis and have various functions in the body. It is necessary to take them in the diet because their deficiency results in decreased formation of protein or protein are not formed as a result protein deficiency may occur. It is concluded that amino acids play important role in our body.

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